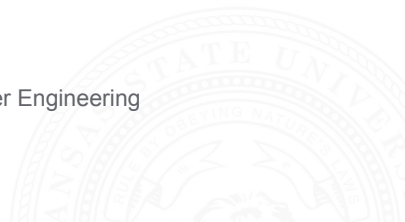


Per Unit System II

From one voltage level to a multi-voltage network

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Lecture outline

- 1 Part 1: Transformer normalization
- 2 Part 2: System bases and equipment-base conversion
- 3 Part 3: Worked Example

Normalize the transformer first, then use its per-unit model to connect different voltage levels.

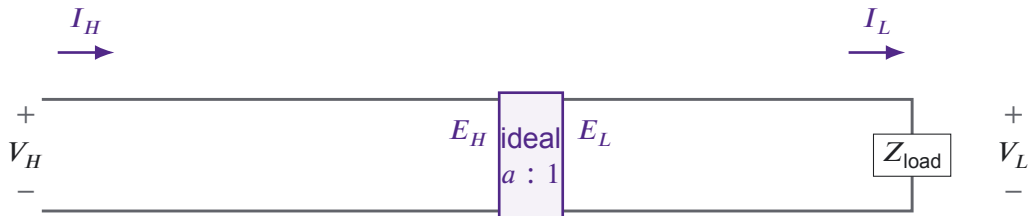
PART 1

Transformer normalization

Corresponding bases on the two sides and the resulting per-unit circuit

The ideal transformer and local bases

The ideal $a : 1$ element establishes the voltage and current ratios



$$V_H = E_H, \quad V_L = E_L$$

$$\text{H side: } (S_b, V_{b,H}), \quad \text{L side: } (S_b, V_{b,L})$$

Start with the ideal transformer. Normalize the H and L sides using their own voltage bases and one common power base.

Transformer per-unit base rules

Single-phase transformer

$$S_{b,H} = S_{b,L} = S_b$$

$$\frac{V_{b,H}}{V_{b,L}} = \frac{V_{H,\text{rated}}}{V_{L,\text{rated}}} = a$$

$$a = \frac{N_H}{N_L}$$

Choose the power base and one voltage base. The other side's voltage base follows from the rated ratio.

The current and impedance bases on each side

Quantity	H-side base	L-side base
Power	S_b	S_b
Voltage	$V_{b,H}$	$V_{b,L} = V_{b,H}/a$
Current	$I_{b,H} = S_b/V_{b,H}$	$I_{b,L} = S_b/V_{b,L}$
Impedance	$Z_{b,H} = V_{b,H}^2/S_b$	$Z_{b,L} = V_{b,L}^2/S_b$

$$I_{b,H} = I_{b,L}/a, \quad Z_{b,H} = a^2 Z_{b,L}$$

The two sides share the power base. Their current and impedance bases generally have different numerical values.

Voltage normalization cancels the turns ratio

Internal winding voltages: $E_H = aE_L$, with $V_{b,H} = aV_{b,L}$

$$E_{H,\text{p.u.}} = \frac{E_H}{V_{b,H}} = \frac{aE_L}{aV_{b,L}} = \frac{E_L}{V_{b,L}} = E_{L,\text{p.u.}}$$

$$E_{H,\text{p.u.}} = E_{L,\text{p.u.}}$$

For the ideal transformer, terminal and internal voltages coincide

$$V_{H,\text{p.u.}} = E_{H,\text{p.u.}} = E_{L,\text{p.u.}} = V_{L,\text{p.u.}}$$

Compatible voltage bases turn the ideal transformer voltage ratio into 1:1 in the per-unit circuit.

Current normalization cancels the inverse ratio

The ideal element alone: exciting current excluded

$$I_H = I_L/a, \quad I_{b,H} = I_{b,L}/a$$

$$I_{H,\text{p.u.}} = \frac{I_H}{I_{b,H}} = \frac{I_L/a}{I_{b,L}/a} = \frac{I_L}{I_{b,L}} = I_{L,\text{p.u.}}$$

$$I_{H,\text{p.u.}} = I_{L,\text{p.u.}}$$

Compatible bases also turn the ideal transformer current ratio into 1:1 in the per-unit circuit.

The ideal per-unit circuit



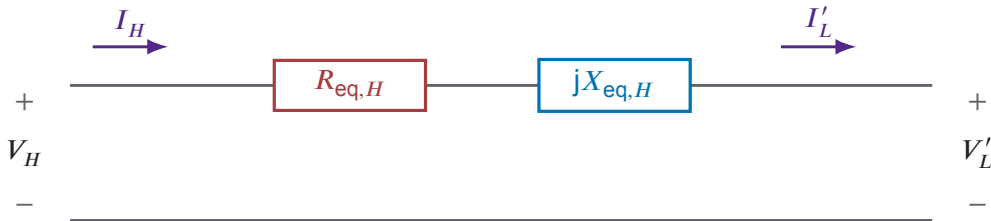
$$E_{H,p.u.} = E_{L,p.u.}, \quad I_{H,p.u.} = I_{L,p.u.}$$

The ideal ratio element is absent from the per-unit circuit.

The drawing represents normalized terminal relationships. Physical voltage and current levels still differ between the two sides.

Approximate circuit referred to H

Neglect excitation; refer both winding impedances to H and combine them



$$R_{eq,H} = R_H + a^2 R_L, \quad X_{eq,H} = X_{\sigma,H} + a^2 X_{\sigma,L}$$

$$V'_L = aV_L, \quad I_H = I'_L = I_L/a$$

This approximate circuit has one series impedance in ohms. Normalizing it gives the series per-unit circuit.

Equivalent impedance before and after normalization

Excitation neglected: combine the two H-referred winding impedances

$$Z_{\text{eq},H} = Z_{w,H} + a^2 Z_{w,L}$$

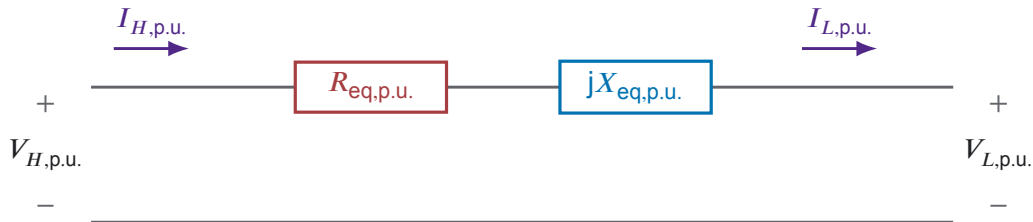
Divide by the H-side base, with $Z_{b,H} = a^2 Z_{b,L}$

$$\begin{aligned} Z_{\text{eq,p.u.}} &= \frac{Z_{\text{eq},H}}{Z_{b,H}} = \frac{Z_{w,H}}{Z_{b,H}} + \frac{a^2 Z_{w,L}}{Z_{b,H}} \\ &= \frac{Z_{w,H}}{Z_{b,H}} + \frac{a^2 Z_{w,L}}{a^2 Z_{b,L}} = \frac{Z_{w,H}}{Z_{b,H}} + \frac{Z_{w,L}}{Z_{b,L}} \end{aligned}$$

$$Z_{\text{eq,p.u.}} = Z_{w,H,\text{p.u.}} + Z_{w,L,\text{p.u.}} = R_{\text{eq,p.u.}} + jX_{\text{eq,p.u.}}$$

Referring first and normalizing gives the same sum as normalizing each winding on its own side. Compatible bases make the per-unit impedances additive.

The series per-unit circuit



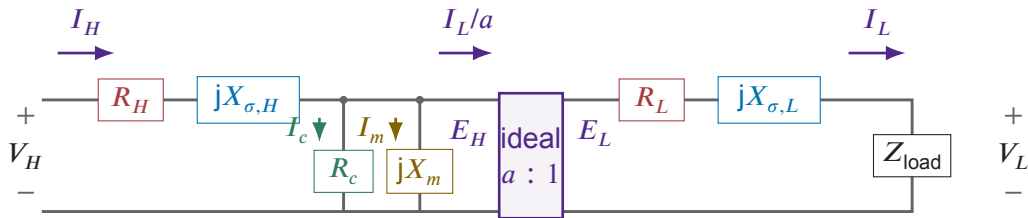
$$I_{H,p.u.} = I_{L,p.u.} = I_{p.u.}$$

$$V_{H,p.u.} - V_{L,p.u.} = I_{p.u.} Z_{eq,p.u.}$$

This is the transformer series model used when exciting current is neglected. Per-unit normalization does not remove its leakage impedance.

The complete physical circuit

Now retain the excitation branch in the complete model



$$Z_{w,H} = R_H + jX_{\sigma,H}, \quad Z_{w,L} = R_L + jX_{\sigma,L}$$

$$E_H = aE_L, \quad I_H = I_0 + I_L/a, \quad I_0 = I_c + I_m$$

Next, normalize the H-side excitation branch to obtain the complete per-unit circuit.

Normalization of the excitation branch

The H-side shunt branch from L11, expressed as admittance

$$Y_0 = G_c - jB_m, \quad G_c = 1/R_c, \quad B_m = 1/X_m$$

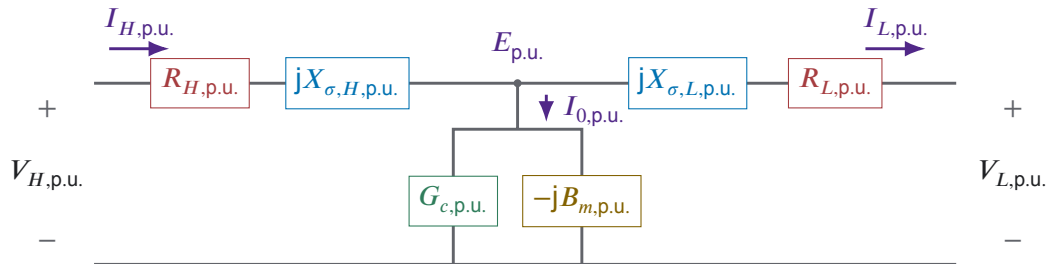
$$Y_{b,H} = 1/Z_{b,H}$$

$$G_{c,p.u.} = G_c/Y_{b,H}, \quad B_{m,p.u.} = B_m/Y_{b,H}$$

$$I_{0,p.u.} = E_{p.u.}(G_{c,p.u.} - jB_{m,p.u.})$$

Normalize the excitation branch on its reference side. Keeping this branch preserves the difference between input and output currents.

The complete per-unit circuit



$$I_{H,p.u.} = I_{L,p.u.} + I_{0,p.u.}$$

$$E_{p.u.} = E_{H,p.u.} = E_{L,p.u.}$$

The ideal ratio is still absent. The excitation branch lies between the two winding impedances, so the windings do not carry the same current.

Transformer normalization: summary

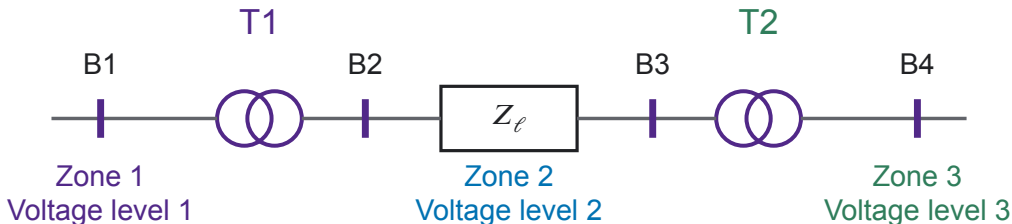
With compatible bases, normalization **removes the ideal turns ratio** from the per-unit circuit, connects the primary and secondary internal ports at the **same per-unit voltage**, and allows circuits at **different physical voltage levels** to form one per-unit network.

PART 2

System bases and equipment-base conversion

One complete per-unit circuit: define compatible bases, convert each component, then combine

The goal: one complete per-unit circuit



Represent T1, the line, and T2 in one per-unit circuit with the ideal transformer ratio elements removed.

$$\text{Target: } Z_{\Sigma, \text{p.u.}} = Z_{T1, \text{p.u.}} + Z_{\ell, \text{p.u.}} + Z_{T2, \text{p.u.}}$$

For the series model, compatible bases let the three impedances combine. First check whether the supplied per-unit data use compatible bases.

The supplied component bases are incompatible

Compare all three components on the Zone 2 side

Given bases	T1	Line	T2
Power base	$S_{T1,\text{rated}}$	S_ℓ	$S_{T2,\text{rated}}$
Voltage base	$V_{T1,2,\text{rated}}$	V_ℓ	$V_{T2,2,\text{rated}}$

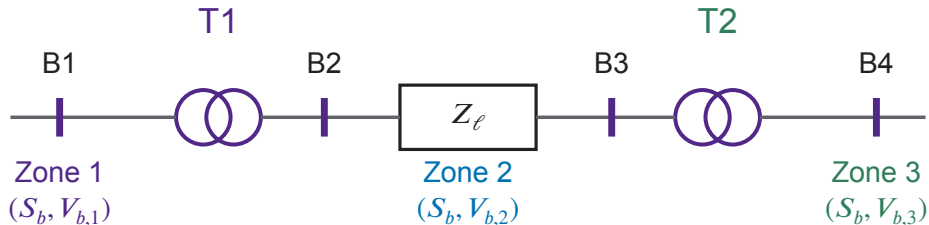
$$I_{b,i} = \frac{S_{b,i}}{V_{b,i}}, \quad Z_{b,i} = \frac{V_{b,i}^2}{S_{b,i}}, \quad Z_{i,\text{p.u.}} = \frac{Z_i}{Z_{b,i}}$$

Independent choices of S_b and V_b can give different I_b and Z_b even for components compared on the same voltage side.

These supplied per-unit impedances cannot be added directly. Equivalent base conversion must first put them on compatible system bases.

System bases for every component

- Choose one power base S_b for the whole system.
- Choose a voltage base per zone, linked by transformer ratios.



Convert any line or transformer data whose bases differ from these targets. Then connect the components in one compatible per-unit circuit.

Change of base

Old: the known bases. New: the desired bases.

$$(S_{b,\text{old}}, V_{b,\text{old}}) \longrightarrow (S_{b,\text{new}}, V_{b,\text{new}})$$

Old and new bases describe the same physical quantity on the same reference side.

$$\text{Single-phase: } I_b = \frac{S_b}{V_b}, \quad Z_b = \frac{V_b^2}{S_b}, \quad Y_b = \frac{S_b}{V_b^2}$$

$$x = x_{\text{p.u.,old}} x_{b,\text{old}} = x_{\text{p.u.,new}} x_{b,\text{new}}$$

$$x_{\text{p.u.,new}} = x_{\text{p.u.,old}} \frac{x_{b,\text{old}}}{x_{b,\text{new}}}$$

The physical quantity x is unchanged. Its new per-unit value follows from its own old-to-new base ratio.

Voltage and current

Use old and new bases on the same physical reference side

$$V_{\text{p.u.,new}} = V_{\text{p.u.,old}} \frac{V_{b,\text{old}}}{V_{b,\text{new}}}, \quad E_{\text{p.u.,new}} = E_{\text{p.u.,old}} \frac{V_{b,\text{old}}}{V_{b,\text{new}}}$$

$$\frac{I_{b,\text{old}}}{I_{b,\text{new}}} = \frac{S_{b,\text{old}}/V_{b,\text{old}}}{S_{b,\text{new}}/V_{b,\text{new}}}$$

$$I_{\text{p.u.,new}} = I_{\text{p.u.,old}} \frac{S_{b,\text{old}}}{S_{b,\text{new}}} \frac{V_{b,\text{new}}}{V_{b,\text{old}}}$$

$$I_{0,\text{p.u.,new}} = I_{0,\text{p.u.,old}} \frac{S_{b,\text{old}}}{S_{b,\text{new}}} \frac{V_{b,\text{new}}}{V_{b,\text{old}}}$$

The same voltage factor applies to V and E ; the same current factor applies to I and I_0 . The physical operating point stays the same.

Impedance

Divide the old impedance base by the new impedance base

$$\frac{Z_{b,old}}{Z_{b,new}} = \left(\frac{V_{b,old}}{V_{b,new}} \right)^2 \frac{S_{b,new}}{S_{b,old}}$$

$$Z_{p.u.,new} = Z_{p.u.,old} \left(\frac{V_{b,old}}{V_{b,new}} \right)^2 \frac{S_{b,new}}{S_{b,old}}$$

$$R_{p.u.,new} = R_{p.u.,old} \left(\frac{V_{b,old}}{V_{b,new}} \right)^2 \frac{S_{b,new}}{S_{b,old}}$$

$$X_{p.u.,new} = X_{p.u.,old} \left(\frac{V_{b,old}}{V_{b,new}} \right)^2 \frac{S_{b,new}}{S_{b,old}}$$

One factor converts resistance, reactance, and complex impedance. Keep the physical reference side unchanged.

Admittance

Admittance $Y = G + jB$ uses the base $Y_b = 1/Z_b$

$$\frac{Y_{b,\text{old}}}{Y_{b,\text{new}}} = \frac{S_{b,\text{old}}}{S_{b,\text{new}}} \left(\frac{V_{b,\text{new}}}{V_{b,\text{old}}} \right)^2$$

$$Y_{\text{p.u.,new}} = Y_{\text{p.u.,old}} \frac{S_{b,\text{old}}}{S_{b,\text{new}}} \left(\frac{V_{b,\text{new}}}{V_{b,\text{old}}} \right)^2$$

$$G_{\text{p.u.,new}} = G_{\text{p.u.,old}} \frac{S_{b,\text{old}}}{S_{b,\text{new}}} \left(\frac{V_{b,\text{new}}}{V_{b,\text{old}}} \right)^2$$

$$B_{\text{p.u.,new}} = B_{\text{p.u.,old}} \frac{S_{b,\text{old}}}{S_{b,\text{new}}} \left(\frac{V_{b,\text{new}}}{V_{b,\text{old}}} \right)^2$$

Conductance and susceptance share the admittance conversion factor. It is the reciprocal of the impedance conversion factor.

Power

Complex, active, and reactive power all use the common power base

$$S = P + jQ, \quad S_{\text{p.u.}} = S/S_b$$

$$S_{\text{p.u.,new}} = S_{\text{p.u.,old}} \frac{S_{b,\text{old}}}{S_{b,\text{new}}}$$

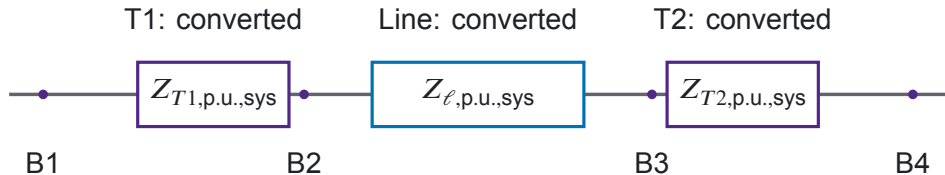
$$P_{\text{p.u.,new}} = P_{\text{p.u.,old}} \frac{S_{b,\text{old}}}{S_{b,\text{new}}}, \quad Q_{\text{p.u.,new}} = Q_{\text{p.u.,old}} \frac{S_{b,\text{old}}}{S_{b,\text{new}}}$$

$$V_{\text{p.u.,new}} I_{\text{p.u.,new}}^* = V_{\text{p.u.,old}} I_{\text{p.u.,old}}^* \frac{S_{b,\text{old}}}{S_{b,\text{new}}}$$

Voltage and current conversion factors give the same power conversion. The physical power remains unchanged.

System application

Series network: nominal ratios, zero phase shift, and no shunts



$$Z_{\Sigma,p.u.} = Z_{T1,p.u.,sys} + Z_{\ell,p.u.,sys} + Z_{T2,p.u.,sys}$$

$$V_{B1,p.u.} - V_{B4,p.u.} = I_{p.u.} Z_{\Sigma,p.u.}$$

Define the system and zone bases, convert every component, then assemble the circuit. Compatible per-unit series impedances can now be added directly.

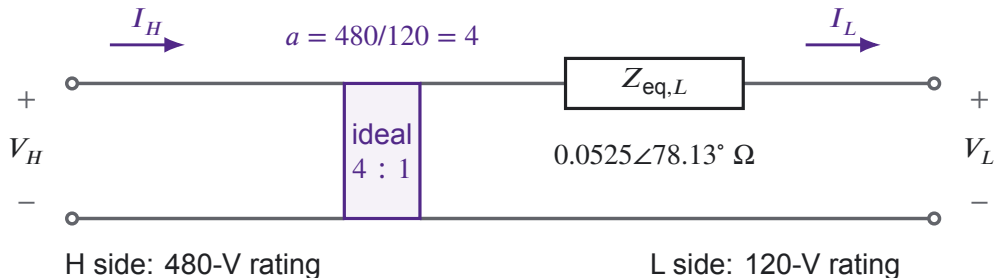
PART 3

Worked example

One transformer, two physical reference sides, one per-unit leakage impedance

Example circuit

Single-phase transformer: 20 kVA, 480/120 V, 60 Hz



Use rated bases to find the per-unit impedance on both sides.

$Z_{eq,L}$ combines both winding leakage impedances, referred to the 120-V side. The series model neglects excitation current.

Step 1: select corresponding voltage bases

$$S_b = 20 \text{ kVA} = 20,000 \text{ VA}$$

$$V_{b,H} = 480 \text{ V}, \quad V_{b,L} = 120 \text{ V}$$

$$a = \frac{480}{120} = 4, \quad \frac{V_{b,H}}{V_{b,L}} = 4$$

$$I_{b,H} = \frac{20,000}{480} = 41.6667 \text{ A}, \quad I_{b,L} = \frac{20,000}{120} = 166.6667 \text{ A}$$

The two voltage bases correspond through the rated ratio. Their derived current bases satisfy $I_{b,H} = I_{b,L}/4$.

Step 2: calculate both impedance bases

$$Z_{b,L} = \frac{120^2}{20,000} = \boxed{0.72 \, \Omega}$$

$$Z_{b,H} = \frac{480^2}{20,000} = \boxed{11.52 \, \Omega}$$

$$\frac{Z_{b,H}}{Z_{b,L}} = \frac{11.52}{0.72} = 16 = a^2$$

Each side has its own impedance base. The ratio of the bases is the same factor used to refer an impedance from L to H.

Step 3: normalize directly on the 120-V side

$$Z_{eq,L} = 0.0525 \angle 78.13^\circ \Omega$$
$$Z_{eq,L,p.u.} = \frac{Z_{eq,L}}{Z_{b,L}} = \frac{0.0525 \angle 78.13^\circ}{0.72}$$

$$Z_{eq,L,p.u.} = 0.0729 \angle 78.13^\circ$$

Use the L-side base because the supplied ohmic impedance is referred to L. Dividing by a positive real base preserves its angle.

Step 4: verify from the 480-V side

Obtain the H-side physical impedance from the supplied L-side data

$$Z_{\text{eq},H} = a^2 Z_{\text{eq},L} = 16(0.0525 \angle 78.13^\circ) = 0.84 \angle 78.13^\circ \Omega$$

Normalize that value using the H-side base

$$Z_{\text{eq},H,\text{p.u.}} = \frac{0.84 \angle 78.13^\circ}{11.52} = \boxed{0.0729 \angle 78.13^\circ}$$

This second route verifies the direct L-side normalization. Both describe the same leakage impedance in per unit.

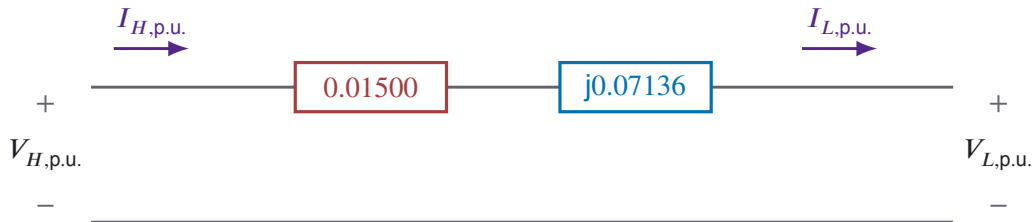
The per-unit result on both sides

	H / winding 1	L / winding 2
Voltage base	480 V	120 V
Impedance base	11.52 Ω	0.72 Ω
Physical leakage impedance	0.84 $\angle$ 78.13° Ω	0.0525 $\angle$ 78.13° Ω
Per-unit leakage impedance	0.0729 $\angle$ 78.13°	0.0729 $\angle$ 78.13°

$$Z_{eq,H,p.u.} = Z_{eq,L,p.u.}$$

Different physical ohmic values and different bases produce the same per-unit transformer parameter.

The transformer series circuit in per unit



$$Z_{eq,p.u.} = 0.0729167 \angle 78.13^\circ \approx 0.01500 + j0.07136$$

$$V_{H,p.u.} - V_{L,p.u.} = I_{p.u.}(0.01500 + j0.07136)$$

With compatible 480/120-V bases, the ideal 4:1 ratio element disappears. The converted resistance and reactance remain in the per-unit circuit.

Recovering quantities on the correct side

$$\begin{aligned}V_H &= 480 V_{H,\text{p.u.}} \text{ V}, & V_L &= 120 V_{L,\text{p.u.}} \text{ V} \\I_H &= 41.6667 I_{H,\text{p.u.}} \text{ A}, & I_L &= 166.6667 I_{L,\text{p.u.}} \text{ A} \\Z_{\text{eq},H} &= 11.52 Z_{\text{eq,p.u.}} \Omega, & Z_{\text{eq},L} &= 0.72 Z_{\text{eq,p.u.}} \Omega\end{aligned}$$

One per-unit circuit represents both voltage levels. Multiplication by the local base recovers the physical value on the selected side.

The same transformer on new system bases

New power and voltage bases

$$S_{b,\text{new}} = 30 \text{ kVA}, \quad V_{b,H,\text{new}} = 600 \text{ V}, \quad V_{b,L,\text{new}} = 150 \text{ V}$$

$$600/150 = 4 = a$$

$$Z_{\text{p.u.,new}} = (0.0729167 \angle 78.13^\circ) \left(\frac{120}{150} \right)^2 \frac{30}{20}$$

$$Z_{\text{p.u.,new}} = 0.0700 \angle 78.13^\circ$$

Both voltage-base and power-base ratios enter the conversion. The physical transformer and its ratings are unchanged.

The new bases preserve the physical impedance

$$Z_{b,L,\text{new}} = \frac{150^2}{30,000} = 0.75 \, \Omega, \quad Z_{b,H,\text{new}} = \frac{600^2}{30,000} = 12 \, \Omega$$

$$(0.0700 \angle 78.13^\circ)(0.75) = 0.0525 \angle 78.13^\circ \, \Omega$$

$$(0.0700 \angle 78.13^\circ)(12) = 0.84 \angle 78.13^\circ \, \Omega$$

Both new descriptions recover the original transformer ohmic values. A base change alters the per-unit number, not the device.



Summary

- 1 Choose one system power base S_b .
- 2 Set one voltage base per zone, linked by the transformer ratios.
- 3 Express every component on its corresponding system bases.

$$Z_{i,\text{p.u.,sys}} = \frac{Z_i}{Z_{b,k}} = Z_{i,\text{p.u.,data}} \frac{Z_{b,i,\text{data}}}{Z_{b,k}}$$

- 4 Connect the normalized models, then combine the applicable series or parallel elements.

Check both power and voltage bases for every component. Convert mismatched data before assembling the system per-unit circuit.